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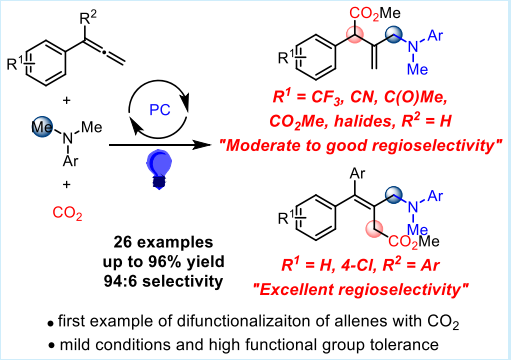
Photoredox-Catalyzed α-Aminoalkylcarboxylation of Allenes with CO_2

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ABSTRACT: The photoredox-catalyzed α-aminoalkylcarboxylation of aryl allenes with CO_2 and *N*,*N*-dimethylanilines is reported for the first time (26 examples, up to 96% yield). In the case of electron-deficient allenes, good regioselectivity was observed (up to 94:6), exclusively generating kinetic products over thermodynamic products. This protocol is a novel synthetic method for highly functionalized β,γ-unsaturated γ-aminobutyric esters.



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γ-Aminobutyric acid is the main inhibitory neurotransmitter in the mammalian central nervous system.[1] For this reason, the synthesis of structurally diverse γ-aminobutyric acid derivatives has been considered important in the pharmaceutical field.[2] In particular, various γ-aminobutyric acids bearing a tertiary carbon center at the β-position such as pregabalin,[3] baclofen,[4] phenibut,[5] and rolipram[6] have been investigated (Figure 1). In this regard, more structurally diverse, β-substituted γ-aminobutyric acid derivatives have been developed to expand the potential bioactive pharmaceutical library.[7]

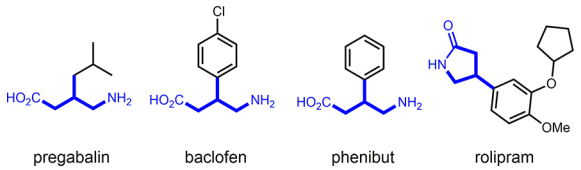


Figure 1. Important β-substituted γ-aminobutyric acids in the pharmaceutical field.

Typical approaches have been developed through the Michael addition of carbonyl compounds to β-substituted nitroalkenes or the conjugate addition of nitromethane to α,β-unsaturated aldehydes.[7a−d] Another approach involves the photocatalytic radical conjugate addition of α,β-unsaturated compounds to amines.[7e−i] Recently, the visible-light-mediated difunctionalization of alkenes with CO_2 and various radical sources to obtain valuable carboxylic acids has received increasing attention.[8]

Among the developed methods, the photoredox-catalyzed α-aminoalkylcarboxylation of electron-deficient alkenes with CO_2 in the presence of 1,2,3,5-tetrakis(carbazol-9-yl)-4,6-dicyanobenzene (4CzIPN) for γ-aminobutyric acid synthesis was independently developed by Wu and Xi (Scheme 1a).[8c,e] Very recently, the photoredox-catalyzed α-aminoalkylcarboxylation of activated alkenes with α-amino acids as a bifunctional reagent for γ-aminobutyric acid synthesis was reported by Yu and Cheng and Sun (Scheme 1a).[8a,h] However, β-substituted γ-aminobutyric acid synthesis with CO_2 remains a great challenge due to the utilization of β-substituted alkenes with high reduction potential.[9] We envisioned that the photocatalytic α-aminoalkylcarboxylation of allenes with CO_2 and amines would serve as an ideal approach to deliver a novel γ-aminobutyric acid through the incorporation of the ethylene moiety at the β-position. Allenes, as special and versatile synthons with structural diversity, have received much attention in organic chemistry.[10]

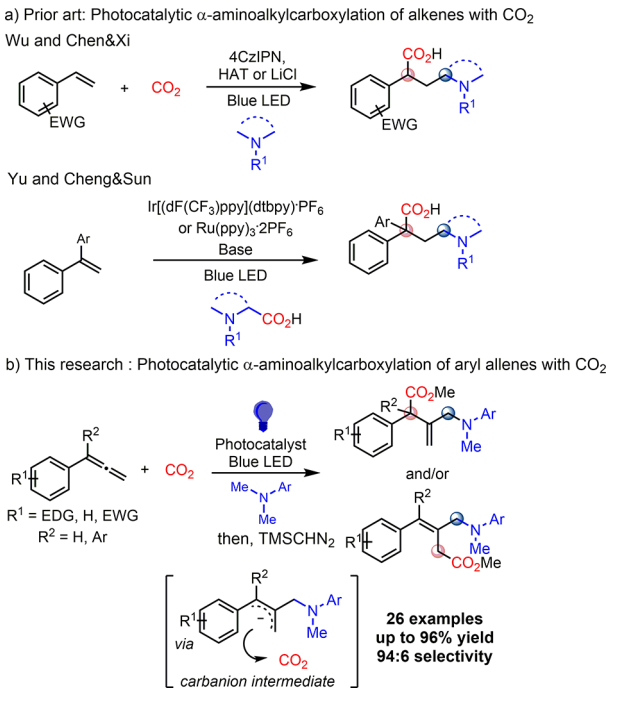
Although various allene difunctionalization methods have been investigated, few studies have been developed on the transition-metal-catalyzed difunctionalization of allenes with CO_2 for unsaturated carboxylic acid derivatives.[11] To the best of our knowledge, there have been no examples demonstrating the photocatalytic difunctionalization of allenes with CO_2 in a redox-neutral fashion. Herein we report the visible-light-mediated α-aminoalkylcarboxylation of aryl allenes with CO_2

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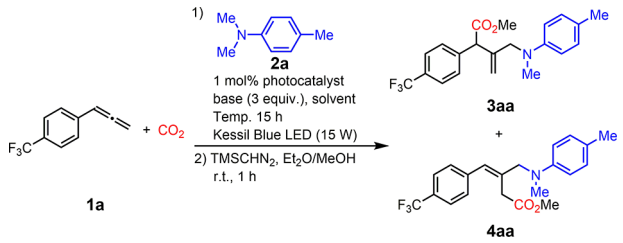
Scheme 1. Sustainable Synthetic Route to γ-Amino Acid Derivatives with CO_2



and amines for β,γ-unsaturated γ-aminobutyric ester synthesis (Scheme 1b). This reaction is the first example of the photoredox-catalyzed difunctionalization of aryl allenes with CO_2. Compared with the photocatalytic difunctionalization of alkenes with CO_2,[8] the photocatalytic difunctionalization of allenes with CO_2 is more challenging due to their regioselectivity control issues.[10,11] A previous report indicated that the combination of a photocatalyst and base was important for regioselective radical addition to 2,3-allenoates.[12] Encouraged by this result, we initially investigated the reaction by using 4-trifluoromethylphenylallene **1a** as a model substrate with *N*,*N*-4-trimethylaniline **2a** as a radical source in the presence of 1 mol % $Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$ when using 1-methyl-2-pyrrolidinone (NMP) and a base under 15 W blue light-emitting diode (LED) irradiation at room temperature with CO_2 at atmospheric pressure (Table 1). We examined the reaction with bases such as $CsOH\cdot H_2O$, NaOAc, and K_2HPO_4 (Table 1, entries 1−3). In base screening, K_2HPO_4 showed the best regioselectivity compared with other bases, although the yield slightly decreased. Next, we tested some other photocatalysts. $Ir(dF(CF_3)ppy)_2(bpy)\cdot PF_6$, $Ir(ppy)_2(bpy)\cdot PF_6$, $Ir(ppy)_3$, 1,2,3,5-tetrakis(carbazol-9-yl)-4,6-dicyanobenzene (4CzIPN), and (2,3,4,6)-3-benzyl-2,4,5,6-tetra(9*H*-carbazol-9yl)benzonitrile (4CzBnBN) gave relatively low regioselectivities and yields (Table 1, entries 4−8). To improve the selectivity of the kinetic product **3aa**, the reaction must proceed at a low temperature. By reacting at 0 °C, the selectivity and yield increased (Table 1, entry 9). We observed that the selectivity significantly increased while reacting at −20 °C, but the yield decreased (Table 1, entry 10). In addition, some other solvents were investigated. *N*,*N*-dimethylformamide (DMF) and *N*,*N*-dimethylacetamide (DMA) were both efficient in terms of the yield and selectivity (Table 1, entries 11 and 12). Finally, no traces of the desired products were detected in the absence of the photocatalyst or light (Table 1, entries 13 and 14).

With these optimized reaction conditions in hand, the aniline substrate scope of α-aminoalkylcarboxylation with **1a** was examined (Scheme 2). Compared with *para*-methyl-

Table 1. Optimization of α-Aminoalkylcarboxylation of Allenes[a]

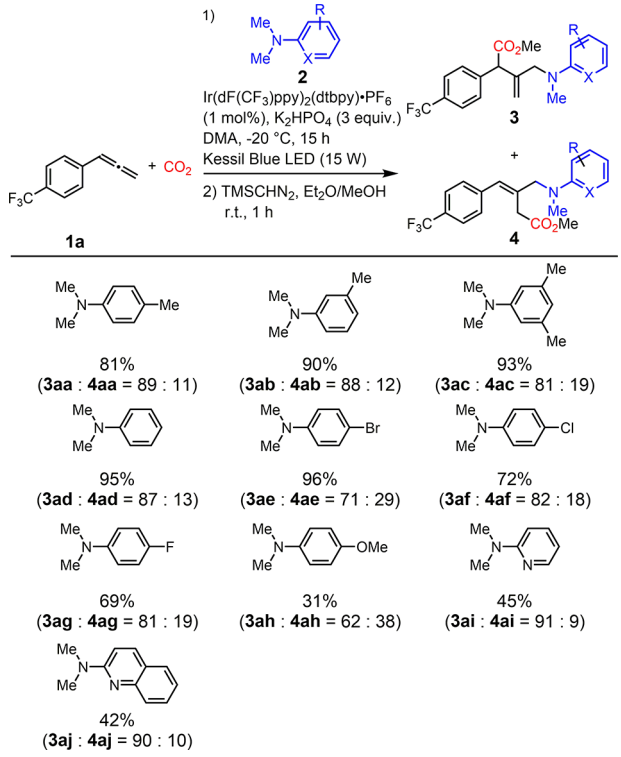


entry	photocatalyst	base	solvent	temp (°C)	yield (%)[b,c]
1	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	$CsOH\cdot H_2O$	NMP	r.t.	79 (66:34)
2	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	NaOAc	NMP	r.t.	66 (73:27)
3	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	NMP	r.t.	63 (77:23)
4	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	NMP	r.t.	60 (66:34)
5	$Ir(ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	NMP	r.t.	64 (76:24)
6	4CzIPN[d]	K_2HPO_4	NMP	r.t.	38 (55:45)
7	4CzBnBN[d]	K_2HPO_4	NMP	r.t.	trace
8	$Ir(ppy)_3$	K_2HPO_4	NMP	r.t.	trace
9	$Ir(dF(CF_3/ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	NMP	0	81 (79:21)
10	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	NMP	−20	64 (87:13)
11	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	DMF	−20	80 (88:12)
12	**$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$**	**K_2HPO_4**	**DMA**	**−20**	**81** (89:11)
13[e]	$Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$	K_2HPO_4	DMA	−20	ND
14[f]		K_2HPO_4	DMA	−20	ND

[a]Conditions: (1) 0.2 mmol **1a**, 0.6 mmol **2a**, 0.6 mmol base, 1 or 3 mol % photocatalyst, solvent (2 mL), CO_2 (1 bar), 15 W blue LED, 15 h. (2) $TMSCHN_2$ 2 M in Et_2O (0.2 mL), Et_2O (8 mL), MeOH (2 mL), 1 h. [b]Yield of **3aa** and **4aa**, isolated yield. ND, not determined [c]Selectivity of **3aa**/**4aa** determined by ^1H NMR. [d]3 mol % [e]Reacted in dark. [f]Without photocatalyst.

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Scheme 2. Scope of α-Aminoalkylcarboxylation of Amines[a,b,c]

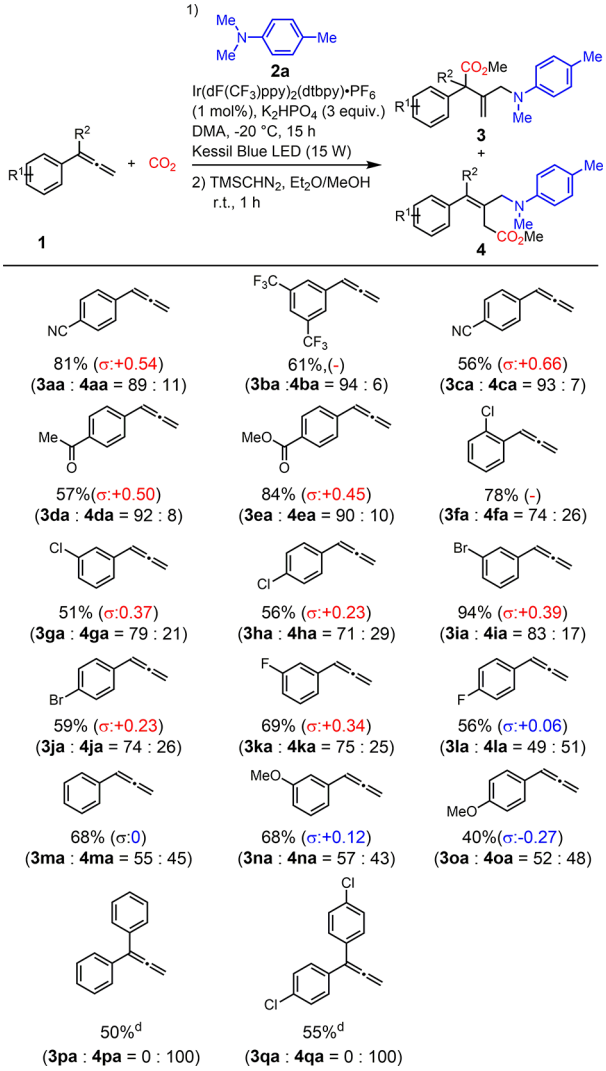


[a]Conditions: (1) 0.2 mmol **1a**, 0.6 mmol **2**, 0.6 mmol K_2HPO_4, 1 mol % $Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$, DMA (2 mL), CO_2 (1 bar), 15 W blue LED, 15 h. (2) $TMSCHN_2$ 2 M in Et_2O (0.2 mL), Et_2O (8 mL), MeOH (2 mL), 1 h. [b]Selectivity determined by ^1H NMR. [c]Isolated yield of **3** and **4**.

substituted *N*,*N*-dimethylaniline (**2a**), *meta*-methyl-substituted *N*,*N*-dimethylanilines (**2b**,**c**) gave higher yields of the desired products, but the regioselectivity slightly decreased (**3**/**4aa**−**ac**). The non- (**2d**), bromo- (**2e**), chloro- (**2f**), fluoro- (**2g**), and methoxy- (**2h**) substituted *N*,*N*-dimethylanilines at the para position of the phenyl ring have an excellent yield and moderate regioselectivity, except for 4-methoxy-*N*,*N*-dimethylaniline (**3**/**4ad**−**ah**). Other heteroaryl amines, such as pyridine and quinoline, successfully gave moderate yields of the desired products (**3**/**4ai**−**aj**). Notably, the utilization of secondary amines and tertiary amines such as 4-dimethylaminopyridine (DMAP), *N*,*N*-dimehtylthiophene-3-amine, potassium (4-methylphenoxy)carbonylaminomethyltrifluoroborate, and *tert*-butyl methyl(*p*-tolyl)carbamate, did not lead to the desired products.[13] (See Scheme S1 in the Supporting Information.)

Next, the reaction scope was studied with aryl allenes **1** and **2a** (Scheme 3). Hammett $\sigma_{m/p}$ values could be in accord with the selectivity of compounds **3** and **4** under the applied reaction conditions.[14] Electron-deficient phenylallenes with positive σ values undergo regioselective α-aminoalkylcarboxylation regardless of the substitution pattern. Phenylallenes bearing highly electron-withdrawing groups such as trifluoromethyl (**1b**), cyano (**1c**), keto carbonyl (**1d**), or ester groups (**1e**) at the para or meta position in the benzene ring were effective at giving the desired kinetic products in good yield with good regioselectivity (**3**/**4ba**−**3**/**4ea**). The introduction of weakly electron-withdrawing substituents, such as chloro

Scheme 3. Scope of α-Aminoalkylcarboxylation of Aryl Allenes[a,b,c]



[a]Reaction conditions: (1) 0.2 mmol **1**, 0.6 mmol **2a**, 0.6 mmol K_2HPO_4, 1 mol % $Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$, DMA (2 mL), CO_2 (1 bar), 15 W blue LED, 15 h, room temperature. (2) $TMSCHN_2$ 2 M in Et_2O (0.2 mL), Et_2O (8 mL), MeOH (2 mL), 1 h. σ = Hammett substituent constants. [b]Selectivity determined by ^1H NMR. [c]Isolated yield of **3** and **4**. [d]Room temperature.

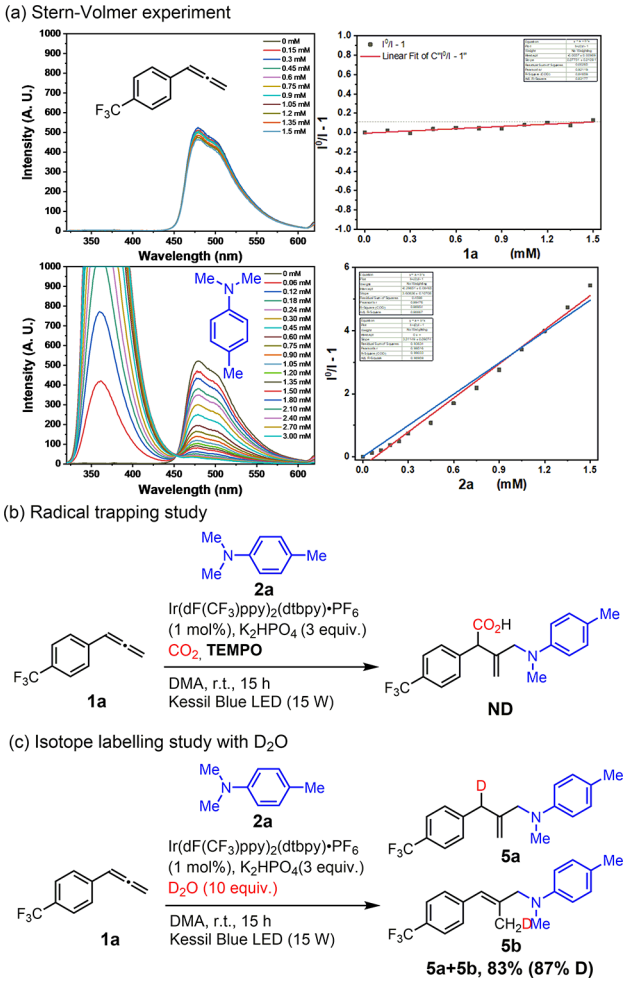
(**1f**−**h**), bromo (**1i**−**j**), and fluoro (**1k**−**l**) groups at the ortho, meta, or para position of the phenyl ring, gave moderate to good yields and regioselectivity, except for 4-fluoro-phenylallene (**3**/**4fa**−**la**). Phenylallene (**1m**) and phenylallene-bearing methoxy groups (**1n**−**o**) at the meta or para position of the phenyl ring exhibited poor regioselectivity (**3**/**4ma**−**oa**). Electron-neutral and electron-donating-group-substituted benzyl radicals with high reduction potentials (−1.43 V vs standard calomel electrode (SCE) for the H benzyl radical analog and −1.75 V vs SCE for the *p*-OMe benzyl radical analog) were difficult to reduce by $Ir(dF(CF_3)ppy)_2(dtbpy)\cdot PF_6$ ($(E_{1/2}Ir(II)/Ir(III))$ = −1.37 V vs SCE in MeCN).[9,15a] Fortunately, it is presumed that allenes are easily reduced via an allyl-benzyl anion intermediate. When using 1,1-diaryl-substituted allenes at room temperature, a thermodynamically favorable product was obtained in moderate yield with excellent regioselectivity (**4pa**−**qa**). The electron density and

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Org. Lett. 2021, 23, 3879−3884

structure of aryl allenes have a strong effect on the regioselectivity. Unfortunately, the utilization of ethyl-2,3-butadienoate did not lead to the desired products. (See Scheme S1 in the Supporting Information.)

To gain insight into the efficiency of the α-aminoalkylcarboxylation of allenes with CO_2, preliminary mechanistic experiments were conducted (Scheme 4). Continuous-

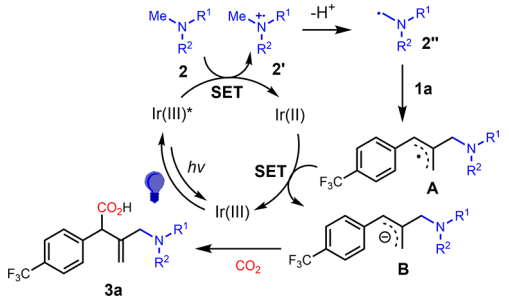
Scheme 4. Mechanistic Studies



light irradiation was an essential factor preventing hydroaminoalkylation.[8c] Stern−Volmer luminescence studies confirm that the light-activated Ir photocatalyst ($(E_{1/2}$Ir(III)*/Ir(II)) = +1.21 V vs SCE in MeCN)[15] was quenched by **2a** (E_{oxp} = +0.74 V vs SCE) rather than **1a** (Scheme 4; see Figures S1 and S2 in the Supporting Information). When the radical scavenger 2,2,6,6-tetramethylpiperdinyloxyl (TEMPO) was used as an additive in the α-aminoalkylcarboxylation, the desired products were not observed. The radical adduct was detected by high-resolution mass spectrometry (HRMS) (Scheme 4 and Figure S3 in the Supporting Information). This result indicates that α-aminoalkylcarboxylation might rely on the radical pathway. The isotope-labeling experiments with D_2O instead of CO_2 showed that benzyl allyl anion species were present in the catalytic cycle (Scheme 4; see Figure S4 in the Supporting Information).

On the basis of the experimental results and the related refs 10, 12, and 16, the proposed mechanism of the regioselective α-aminoalkylcarboxylation to aryl allenes is outlined in Scheme 5. The α-aminomethyl radical (**2′′**), which is produced by a photoexcited photocatalyst and base, undergoes addition to the

Scheme 5. Proposed Mechanism



central carbon of the allenes (**1a**) to give a radical delocalized intermediate (**A**). A subsequent single-electron transfer (SET) between A and the reduced photocatalyst Ir(II) gives the delocalized allyl-benzyl carbanion. The carbanion intermediates (**B**) attack CO_2 as nucleophiles, affording the expected β,γ-unsaturated γ-aminobutyric esters (**3a**) after treatment with $TMSCHN_2$. The photocatalyst was regenerated by SET in the catalytic cycle. A gram-scale reaction was readily conducted. (See Scheme S4 in the Supporting Information.) To demonstrate the utility of this reaction, the unsaturated γ-aminobutyric ester **3aa** was reduced by lithium aluminum hydride (LAH) to form δ-amino alcohol **6**. (See Scheme S5 in the Supporting Information.)

In summary, we demonstrated the photoredox-catalyzed intermolecular α-aminoalkylcarboxylation of aryl allenes with CO_2 and amines in a redox-neutral fashion for the first time. Twenty-six examples of various aryl allenes and amines were demonstrated to afford good isolated yields (up to 96%). Electron-deficient substituted aryl allenes show great regioselectivity. This sustainable process is the general synthetic method of unsaturated γ-aminobutyric esters. α-Aminoalkylcarboxylation has the advantages of step and atom economy and mild conditions. The application and detailed reaction mechanism are currently ongoing.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c01011.

Detailed experimental procedures, mechanistic studies, and spectral data for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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